



Russia launches UK satellites

36 UK Telecommunications Satellites (Soyuz rockets) have been launched from Russia's Far East. <http://dgit.in/apr21-53>

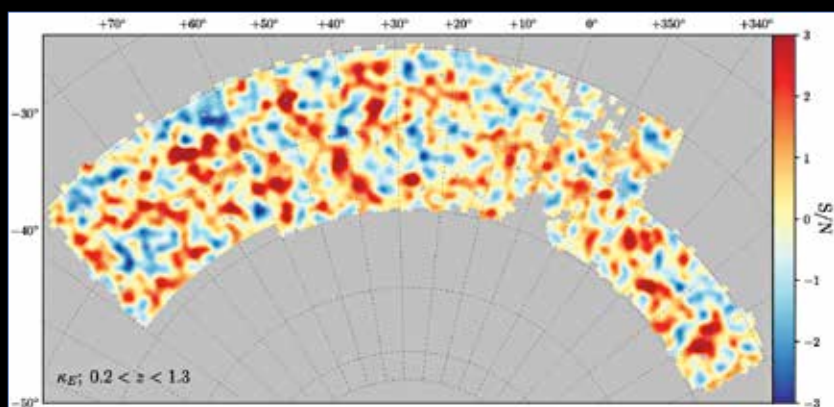


Selfie with Mont Mercou

NASA's Curiosity Mars rover photographs the peculiar rock formations on Mars called Mont Mercou. <http://dgit.in/apr21-54>

distance to the object. The DES has produced the most comprehensive survey of distant galaxies, with 300 million objects studied. The data is a catalogue of over 700 million astronomical objects, in addition to the 400 million found in the project's first run. The project wound up in January this year, having collected 50 terrabytes of data in its second run, which is stored at the US National Centre for Supercomputing Applications (NCSA), at the University of Illinois. Here the NCSA processes the data using the Blue Waters supercomputer at NCSA, the Illinois Campus cluster, as well as systems from Fermilab. This calibrated data is released to the public and the scientific community for further research. It takes about four months to prepare a years worth of data. While the observation has wound up, the project is expected to continue to release data for specific purposes throughout the 2020s.

DES Director Rich Kron of Fermilab and the University of Chicago says, "This is a momentous milestone. For six years, the Dark Energy Survey collaboration took pictures of distant celestial objects in the night sky. Now, after carefully checking the quality and calibration of the images captured by the Dark Energy Camera, we are releasing this second batch of data to the public. We invite professional



A mass map of galaxy densities from DES data, the red regions are more massive

and amateur scientists alike to dig into what we consider a rich mine of gems waiting to be discovered." Keith Bechtol, assistant professor in physics at UW–Madison served as the DES science release coordinator since 2017, and said, "most of the nearly 700 million objects visible in DES DR2 images had never been seen by humans before the past few years. If you take a moment to look at even a small patch of sky in the DES images, you can see asteroids of our Solar System, stars out to the edge of the Milky Way, and distant galaxies as they were billions of years ago. We look forward to see how our colleagues use this enormous new dataset for research and education."

Bhuvnesh Jain, a professor in Physics and Astronomy at Penn State university says, "DES has imaged millions of galaxies at dif-

ferent stages of cosmic evolution. The release of the full DES data is a big deal for the astronomy community since it is one of the largest and best calibrated datasets of stars and galaxies." Researchers from Penn State contributed to the project by refining the calibration techniques, and developing the algorithms to automate the image processing.

Many studies and papers are expected to be based on this data dump, and researchers are expected to keep processing the data for years. Previously data from the DES has been used to produce the largest map of dark matter and dark energy in the universe. The new data is being utilised to create an updated map, one that will be three times larger, and provide a more accurate picture of the distribution of dark matter and dark energy in the universe. Gary Bernstein, one of the scientists from Penn state who helped planned the observation campaign and calibrated the camera, says, "In the next two years, DES will produce the most precise measurements of how cosmic structures have grown via gravity. These will in turn provide stringent tests of the standard model of cosmology—that the universe is composed primarily of dark matter and dark energy, acting as a 'cosmological constant' that accelerates the expansion of the universe." 



The Dark Energy Camera